Rationale-Augmented Training for Targeted Hate Speech Detection in Vietnamese Social Media: A Preliminary Investigation

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***Abstract*—We adapt the HARE rationale-augmented training framework to Vietnamese targeted hate speech detection on the ViTHSD dataset (10,001 comments, 11 classes in a multi-label setting). A two-stage QLoRA pipeline fine-tunes Qwen2.5-3B-Instruct: Stage 1 trains a multi-label classifier on 7,540 oversampled examples; Stage 2 continues on 1,221 rationale-augmented training tuples from LLM-generated, human-reviewed implied statements, providing training-time semantic supervision that encodes target identification, harm-mechanism reasoning, and severity justification. Multi-seed evaluation (seeds 42, 123, 456) shows HARE achieves the highest F1-Macro (35.93±0.63% vs. next-best 32.75±0.53%) and Recall-Micro (62.15±0.75% vs. 59.79±1.55%) among all models, while Vanilla Stage 1-only leads aggregate F1-Micro (60.04±0.40% vs. 57.08±1.08%). This aggregate advantage is confounded by differential parse-failure rates (∼36% HARE vs. ∼59% Vanilla): a paired bootstrap on parseable outputs reverses the ranking (HARE 46.71% vs. Vanilla 44.58%, p=0.004). Per-label analysis reveals HARE outperforms Vanilla on 6 of 8 non-religion labels, with disproportionate Offensive-level gains (+6.84 avg F1) alongside modest Hate-level improvement (+2.77 avg F1). Precision–recall decomposition shows these gains are recall-driven. Matched-budget ablation, confirmed across two seeds, provides directional evidence that rationale semantic structure contributes to these improvements.**

***Index Terms*—hate speech detection, Vietnamese NLP, rationale-augmented training, QLoRA, ViTHSD, Qwen2.5**

I. Introduction

Vietnamese social media platforms—Facebook, YouTube, and TikTok—have become primary channels for public discourse while simultaneously enabling a surge in toxic and hateful content. The Vietnamese language presents unique moderation challenges: it employs a syllabic-based script with ambiguous word boundaries [16], is tonal, meaning is context-dependent, and users frequently encode hate in cultural slang (e.g., *"ba que," "bò đỏ"*), teencode, sarcasm, and wordplay that evade keyword-based filters [17], [18]. Three compounding limitations affect existing automated approaches: (1) *lack of implicit-meaning modeling*—classifiers operate on surface tokens without capturing the underlying stereotypes or harm mechanisms that distinguish Offensive from Hate content [12]; (2) *inability to detect implicit hate*—surface-level models miss attacks expressed through cultural metaphor or sarcasm [14]; (3) *neglect of attack targets*—prior work focuses on severity while overlooking who is attacked—an individual, a group, or a political entity—information essential for calibrating moderation responses [13]. A further difficulty is the *Offensive–Hate boundary*: implicit expressions often occupy the ambiguous space between insult and dehumanizing attack, causing boundary instability even when target identification is correct.

We address these challenges on ViTHSD [2], a dataset of 10,001 Vietnamese social media comments annotated with composite *[Target#Level]* labels, where Target ∈ {Individuals, Groups, Religion, Race, Politics} and Level ∈ {Offensive, Hate}, plus a Normal label—yielding 11 classes in a multi-label setting where each comment may carry multiple target-level pairs. We adapt the HARE framework [9] to this Vietnamese targeted task by generating 1,744 Chain-of-Thought (CoT) rationale tuples via Gemini 3.1 [6] and training Qwen2.5-3B-Instruct via two-stage QLoRA [11].

This paper makes two focused contributions:

- **Rationale-Augmented Training for Vietnamese Targeted Hate Speech with Parse-Failure Analysis.** We present the first empirical study of training-time rationale supervision on Vietnamese targeted hate speech (ViTHSD). HARE achieves the best F1-Macro and Recall-Micro across all models and the best parseable-only F1-Micro (p=0.004). We show that aggregate F1-Micro rankings are confounded by differential parse-failure rates (∼36% HARE vs. ∼59% Vanilla), and that naïve constrained decoding catastrophically fails, motivating retraining-based solutions.Rationale supervision operates exclusively at training time as semantic regularization; the deployed model produces label-only predictions without inference-time explanations, so we claim improved label quality rather than end-to-end explainability.
- **Hate/Offensive Boundary Redistribution.** Validated across three seeds, Stage 2 rationale training consistently shifts predictions toward Offensive-level labels (+6.84

avg F1) with modest Hate-level gains (+2.77 avg F1)—an asymmetric improvement invisible from aggregate metrics and not previously characterized in Vietnamese hate speech literature.

We emphasize that these results constitute exploratory evidence from a limited-seed evaluation on a single dataset. We do not claim rationale semantics definitively cause better detection; rather, we characterize a previously undocumented boundary phenomenon and provide initial ablation evidence for the underlying mechanism.

Sections II–IV review related work, methodology, and rationale generation. Sections V–VII present experiments, results, and ablation studies. Section VIII concludes.

II. Related Work

Vietnamese NLP and Hate Speech. Pre-trained Transformer models for Vietnamese include PhoBERT [3], PhoGPT [20], and multilingual Qwen2.5 [4]. ViHSD [1] (33,400 comments; Clean/Offensive/Hate) provided the first large-scale Vietnamese hate speech dataset; ViTHSD [2] adds five target categories with multi-label severity pairs, reporting ViSoBERT at F1-Micro 51.12%. ViHOS [21] provides span-level rationale annotation; our approach differs by using document-level CoT rationales applied only during training.

Explainable Detection and Parameter-Efficient Fine-Tuning. The HARE framework [9] uses LLM-generated step-by-step rationales to improve both interpretability and classification. HateXplain [12] and Social Bias Frames [13] motivate target-aware explainable detection. Chain-of-Thought prompting [10] decomposes complex reasoning into interpretable chains. QLoRA [11] enables adapting large models on consumer hardware, making CoT-enhanced training feasible in resource-constrained settings.

Offensive–Hate Boundary. Davidson et al. [22] first documented systematic Offensive→Hate confusion; OLID [23] and SemEval-2019 Task 5 [19] report analogous inter-level instability. ElSherief et al. [14] show implicit hate occupies a low-confidence continuum; Sap et al. [13] demonstrate that perceived severity depends on who is attacked. In Vietnamese, cultural slang and indirect reference compound this challenge. Crucially, no prior study has tested whether training-time rationale supervision can systematically shift this boundary; our per-label analysis provides the first such evidence in a Vietnamese multi-label setting.

III. Methodology

Our approach follows a two-phase pipeline (Fig. 1): (1) **Annotation**—generating rationales via a commercial LLM; and (2) **Training**—fine-tuning Qwen2.5-3B-Instruct in two stages. Unlike the original HARE [9], which extracts rationales at both training and inference time, our adaptation employs rationale supervision exclusively during training; inference produces only label predictions.

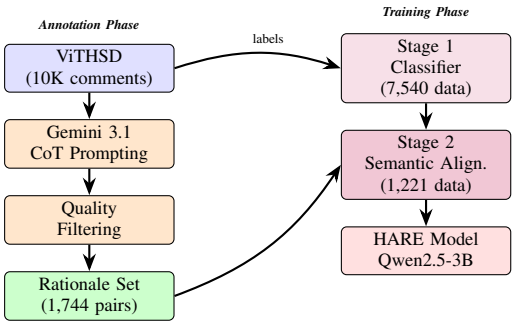

Figure 1. Two-phase pipeline: annotation (left) and two-stage training (right).

A. Model Selection

We select **Qwen2.5-3B-Instruct** [4]: (1) at 3B parameters it fits on a single T4 GPU (16 GB) under 4-bit quantization; (2) it performs competitively on Vietnamese benchmarks among sub-10B models vs. Llama 2-7B [7] and Gemma-7B [8]; and (3) its instruction-following capability supports structured output natively.

B. Two-Stage QLoRA Fine-Tuning

Both stages use QLoRA [11] with 4-bit NF4 quantization, LoRA [15] rank r=8, α=16, dropout 0.1, targeting all attention and feed-forward projection layers. Input uses the Qwen chat template.

Stage 1 (Classification): The model learns to predict multi-label *[Target#Level]* from the ViTHSD training set. We oversample labels with fewer than 200 training examples via random sampling with replacement—540 additional samples concentrated in Politics, Race, and Religion—yielding 7,540 training examples used by all baselines. A post-processing step enforces one-level-per-target by preferring Hate over Offensive. Hyperparameters: 2 epochs, batch 4, lr 2×10^{-4}, max length 512.

Stage 2 (Rationale-Augmented Continuation): Starting from the Stage 1 checkpoint, training continues on 1,221 rationale-enriched training tuples—the training subset of the 1,744 annotated pairs (Sec. IV). Inputs are augmented with implied statement and rationale fields, teaching the model to associate implicit expressions with their true semantic intent. Hyperparameters identical to Stage 1. The 1,221 samples are a subset of the original training set; some comments are seen twice (plain in Stage 1, rationale-enriched in Stage 2), which may contribute to memorization effects.

Attribution limits. Stage 2 changes both training content and optimization trajectory; matched-budget ablation controls are reported in Sec. VII.

C. Inference

The model generates free-text output (max 50 tokens) parsed via string matching against the 11-label vocabulary. All detected label strings are accepted without a confidence threshold (multi-label output accepts any subset of the 11 labels detected in the generated text), subject to a one-level-per-target

constraint that prefers Hate over Offensive when both are detected for the same target. If parsing fails—at rates that vary substantially by model and training stage (Sec. VI-B)—the comment defaults to *Normal*, inflating Normal-class precision and suppressing minority-label recall.

IV. Rationale Generation and Quality

We employ Gemini 3.1 [6] as the automated annotator. Through four prompt iterations—progressing from a base Role-Task-Output template (mean annotator score 40/100) through psychologist-persona few-shot (67) and expert-analyst constrained CoT (89) to the adopted design (93)—we converged on a structured two-prompt pipeline. The final design comprises: (a) an *implied-statement prompt* that extracts noun-phrase stereotypes constrained to seven predefined categories (competence, morality, gender, ethnicity, class, religion/politics, appearance) with explicit negative examples to prevent speaker-action paraphrases; and (b) a *rationale prompt* enforcing a strict four-step CoT structure per comment: (1) *Nhận diện ngôn từ*—identify offensive language cues; (2) *Cơ chế gây hại*—explain the harm mechanism; (3) *Liên kết implied statement*—connect to the extracted stereotype; (4) *Phân biệt mức độ*—justify Hate vs. Offensive distinction. Outputs are enforced as structured JSON; scores reflect mean 4-annotator holistic ratings on label consistency and cultural plausibility (0–100 scale).

Quality filtering. LLM-generated implied statements were produced for all 7,528 hate/offensive training instances and reviewed by four annotators (Vietnamese university students, ages 20–22, Ho Chi Minh City; regional diversity not controlled) for cultural appropriateness and label alignment. No formal IAA metric (Cohen's κ or Krippendorff's α) was computed; this is an acknowledged limitation. All five team members participated in consensus-based verification of $\sim$300 stratified tuples; disagreements were resolved through discussion. A 50-sample formal audit protocol with dual-reviewer Likert scoring (relevance, faithfulness, completeness) was designed but not completed before submission. Remaining pairs passed an automatic consistency check: re-fed into Gemini for label re-prediction, with mismatches discarded. Same-family verification is a limitation (Sec. VII). The final set contains **1,744 tuples** across all data splits (1,221 training, 321 test, 202 development; selection rate $\approx$23%); 5,784 (76.8%) were rejected, concentrated on ambiguous boundary cases, shorter posts, and minority labels. Table I reports label distribution: Hate-level labels constitute 60.5% of label instances (1,582/2,617), confirming strong bias toward high-confidence Hate examples—a selection effect likely contributing to boundary redistribution (Sec. VI-B).

Selection bias analysis. The automatic consistency filter produces a Hate/Offensive ratio reversal: ViTHSD training data is Offensive-dominant (66.3% of label instances), whereas the rationale set is Hate-dominant (60.5%, Δ=+26.8pp). Per-label shifts are largest for Individuals#Hate ($+$14.5pp) and Individuals#Offensive ($-$19.0pp). This inversion arises because dehumanizing Hate-level language provides clearer reasoning anchors for the LLM, while ambiguous Offensive cases are disproportionately rejected. The reversal directly connects to the boundary redistribution (Sec. VI-B): Stage 2 training on a Hate-biased subset shifts the decision boundary, improving Offensive detection while slightly degrading Hate-level recall.

Table I
Label distribution of the 1,744 accepted rationale tuples. Counts exceed 1,744 due to 718 multi-label comments.

Label	Count	%*
Individuals#Hate	878	50.3
Individuals#Offensive	566	32.5
Groups#Hate	451	25.9
Groups#Offensive	319	18.3
Politics#Hate	137	7.9
Race#Hate	110	6.3
Race#Offensive	88	5.0
Politics#Offensive	52	3.0
Religion#Offensive	10	0.6
Religion#Hate	6	0.3
Hate-level total	1,582	60.5†
Offensive-level total	1,035	39.5†

*% of 1,744 comments; sums >100 due to multi-label.
†% of 2,617 total label instances.

V. Experimental Setup

Dataset and Metrics. ViTHSD [2] contains 10,001 Vietnamese comments with composite *[Target#Level]* labels across five targets (Individuals $\sim$60%, Groups $\sim$33%, Race $\sim$5%, Politics $\sim$4%, Religion $<$0.3%) and two severity levels per target, yielding 11 classes. All models train on the 7,540-sample oversampled training split and evaluate on the 1,800-sample test set. We report F1-Micro, F1-Macro, F1-Samples, Precision-Micro, and Recall-Micro via scikit-learn. HARE and Vanilla are evaluated under three random seeds (42, 123, 456); baselines are single-run. All experiments run on NVIDIA T4 GPUs (16 GB, Kaggle; $\sim$4.5 h per seed).[1] F1-Macro is our primary metric as it weights all 11 classes equally, making it more informative than F1-Micro for highly imbalanced settings.

Baselines (all retrained under the same oversampled 7,540-sample protocol; raw text input, no word segmentation):

1) **BiGRU-LSTM-CNN**: Hybrid sequential model; batch 16, 3 epochs, lr 10^{-3}, max length 256.
2) **PhoBERT-base** [3]: Encoder-only; batch 16, 3 epochs, lr 2×10^{-5}. *Note*: pretrained on word-segmented text; unsegmented input is a preprocessing disadvantage not shared by Qwen2.5 or Flan-T5.
3) **Flan-T5-base** [5]: Encoder-decoder; batch 8, 3 epochs, lr 3×10^{-5}, max output 64 tokens.
4) **Qwen2.5-3B Vanilla**: Stage 1 QLoRA only (no rationale data).
5) **HARE (this work)**: Stage 1 + Stage 2 rationale continuation.

[1]Code, rationale tuples (1,744 pairs), and LoRA adapters will be released upon acceptance; a browser-based demo is available upon request.

Table II
PER-SEED PARSE-FAILURE RATES AND PARSEABLE-ONLY F1-MICRO (%). BOOTSTRAP p-VALUES FROM 10,000-ITERATION PAIRED TESTS.

	Fail Rate (%)		Parse-only F1		
Seed	**HARE**	**Van.**	**HARE**	**Van.**	p
42	33	57	47.2	44.6	0.004
123	35	60	45.8	44.9	0.377
456	40	60	47.1	44.3	0.019
Pooled	36	59	46.7	44.6	0.004

Table III
MULTI-LABEL PERFORMANCE ON VITHSD TEST SET (%). HARE/VANILLA: MEAN±STD OVER SEEDS {42,123,456}. BASELINES (†): SINGLE-RUN. BOLD = BEST PER METRIC.

Metric	**HARE** ($\mu\pm\sigma$)	**Vanilla** ($\mu\pm\sigma$)	**BiGRU-LSTM**†	**PhoBERT**†	**Flan-T5**†
F1-Samples	60.67±0.90	**63.77**±0.68	58.49	58.01	50.74
F1-Micro	57.08±1.08	**60.04**±0.40	51.91	54.12	46.84
F1-Macro	**35.93**±0.63	32.75±0.53	31.90	25.86	13.11
Prec-Micro	52.82±2.18	**60.37**±2.04	41.98	56.20	48.10
Rec-Micro	62.15±0.75	59.79±1.55	**67.98**	53.10	45.20

Table IV
PER-LABEL F1 (%) ON VITHSD TEST SET. HARE/VANILLA: MEAN±STD OVER 3 SEEDS. Δ: HARE−VANILLA. BASELINES: SINGLE-RUN. †: <40 TEST SAMPLES. "–" = F1 = 0. BOLD = BEST MEAN PER ROW.

Label	**HARE** ($\mu\pm\sigma$)	**Vanilla** ($\mu\pm\sigma$)	Δ	**BiGRU-LSTM**†	**PhoBERT**†	**Flan-T5**†
Normal	75.7±0.5	**79.2**±0.9	−3.5	74.90	75.31	67.32
Individuals#Hate	**61.7**±1.0	60.9±1.0	+0.8	58.01	53.65	35.83
Individuals#Offensive	**42.0**±1.3	28.7±6.5	+13.4	38.60	7.73	–
Groups#Hate	51.9±2.5	**55.0**±1.1	−3.2	43.84	46.03	34.93
Groups#Offensive	**33.4**±2.7	20.0±5.7	+13.4	27.98	–	–
Religion#Hate†	–	–	–	–	–	–
Religion#Offensive†	–	–	–	–	–	–
Politics#Hate	51.4±2.6	**51.6**±1.1	−0.3	43.60	29.13	2.99
Politics#Offensive	4.6±4.0	15.2±5.8	−10.6	15.19	**21.18**	0.00
Race#Hate	**48.4**±5.3	34.7±16.4	+13.7	29.91	28.57	3.12
Race#Offensive	**26.1**±4.2	14.9±4.5	+11.2	18.89	22.92	–

VI. RESULTS AND ANALYSIS

This section presents multi-seed aggregate performance, per-label boundary analysis, and error characterization for the ViTHSD test set (1,800 samples).

A. Overall Performance

Table III presents multi-label performance (HARE and Vanilla averaged over three seeds). Under standard free-text evaluation, Vanilla achieves higher F1-Micro (60.04±0.40% vs. 57.08±1.08%) and Precision-Micro (60.37±2.04% vs. 52.82±2.18%), while HARE leads on Recall-Micro (62.15±0.75% vs. 59.79±1.55%) and F1-Macro (35.93±0.63% vs. 32.75±0.53%). HARE thus exhibits a high-recall/lower-precision profile versus Vanilla's more balanced output. BiGRU-LSTM-CNN achieves the highest Recall-Micro (67.98%) among single-run baselines at the cost of precision.[2]

Parse-failure confound. Free-text generation with string-matching produces unparseable outputs at substantially different rates: HARE averages ∼36% failures (range 33–40% across seeds) versus Vanilla's ∼59% (range 57–60%). All unparseable outputs default to *Normal*, inflating Normal-class precision and suppressing minority-label recall—disproportionately penalizing the model with *fewer* failures. A paired bootstrap test (10,000 iterations) on parseable outputs only—pooled across three seeds—reverses the ranking: HARE 46.71% vs. Vanilla 44.58% F1-Micro (p=0.004). Per-seed results are consistent: seed 42 (p=0.004), seed 456 (p=0.019), seed 123 (p=0.377, not significant individually).

B. Per-Label Analysis and Boundary Redistribution

Table IV presents seed-averaged per-label F1 scores. HARE outperforms Vanilla on most non-religion labels: the three high-frequency Offensive labels show large gains (Individuals#Offensive +13.36, Groups#Offensive +13.43, Race#Offensive +11.16), and Race#Hate reverses from parity to a HARE lead (+13.68). Individuals#Hate shows a modest gain (+0.82). Vanilla retains advantages on Groups#Hate (+3.16) and Politics#Offensive (+10.59); Politics#Hate is essentially tied (Δ<0.3). The **Boundary Redistribution Index**—average HARE–Vanilla delta across non-religion labels—quantifies the asymmetry: $\Delta\overline{\text{Offens.}} = +6.84$ vs. $\Delta\overline{\text{Hate}} = +2.77$, with Offensive gains consistent across all three seeds. Possible mechanisms are discussed in Sec. VII.

Data scarcity. Religion scores 0% F1 across most models in all seeds (<30 test samples; grayed rows in Table IV). Race and Politics labels remain low-resource; conclusions for these categories are provisional. Accordingly, Race and Politics F1 values in Table IV should be interpreted as low-power point estimates with high variance (Race#Hate σ=5.3, Politics#Offensive σ=4.0–5.8) rather than reliable performance indicators. Statistical conclusions from this paper apply primarily to Individuals and Groups targets, which together represent ∼93% of test instances.

Precision–Recall decomposition. Table V decomposes the boundary redistribution into precision and recall components. HARE's Offensive gains are overwhelmingly *recall-driven*: Individuals#Offensive recall jumps from 21.5% to 50.5% (+29.0pp) while precision drops −8.5pp; Groups#Offensive shows the same pattern (+31.2/−5.1pp). This asymmetry suggests that rationale training primarily lowers the prediction threshold for Offensive-level labels rather than improving discrimination per se.

Post-processing interaction. We verified the one-level-per-target rule has zero effect: reversing the preference (Offensive over Hate) produces identical outputs across all models and seeds, confirming both levels are never co-predicted for the same target.

[2] All retrained baselines exceed the uncontrolled ViSoBERT reference (F1-Micro 51.12% [2]); this gap primarily reflects the 540-sample oversampling protocol.

Table V
PER-LABEL PRECISION (P) AND RECALL (R) DECOMPOSITION (%) FOR HARE VS. VANILLA, MEAN OVER 3 SEEDS. Δ: HARE−VANILLA.

	HARE		Vanilla		Δ (H−V)	
Label	**P**	**R**	**P**	**R**	**ΔP**	**ΔR**
Normal	86.8	67.3	73.4	86.0	+13.4	−18.7
Indiv#Hate	54.2	71.9	56.4	66.7	−2.3	+5.1
Indiv#Offensive	36.2	50.5	44.7	21.5	−8.5	**+29.0**
Groups#Hate	44.5	62.9	57.6	54.2	−13.2	+8.6
Groups#Offensive	26.4	46.1	31.5	15.0	−5.1	**+31.2**

Post: *"Gọi mấy thằng CA là phải dí thẳng tiền cho chúng nó, ko là đến già chúng nó cũng chả quan tâm."*
Gold labels: *Indiv#Hate, Groups#Hate.*
Implied: Công an tham nhũng (*"Police are corrupt"*).
Stage 2 Rationale: (1) *"Thằng CA"* uses a derogatory pronoun (*Indiv* target); (2) *"dí thẳng tiền"* implies bribery (*Groups*); (3) repeated *"chúng nó"* expresses dehumanising contempt; (4) hostile tone consistent with *Hate*-level.

Figure 2. Stage 2 rationale example connecting implicit expressions to Hate-level targets (from ViTHSD training set).

C. Error Taxonomy

Manual inspection reveals four failure modes: (1) **Parse failures** (HARE: 36%, Vanilla: 59%)—unparseable output defaulting to Normal; (2) **Boundary confusion**—correct target, wrong severity (∼18% HARE vs. ∼12% Vanilla errors); (3) **Multi-target omission**—HARE recovers secondary targets in ∼41% of multi-label cases vs. Vanilla's ∼34%; (4) **Implicit hate via teencode/sarcasm**—no-diacritics Vietnamese remains systematically harder for both models.

VII. ABLATION STUDY AND DISCUSSION

Matched-budget ablation controls. To disentangle rationale semantic content from extra gradient steps and training-distribution shifts, we ran two matched-budget controls on the same 1,221-sample Stage 2 training subset (single seed, seed 42): (a) *Shuffled-Rationale*—rationale steps randomly permuted within each example, preserving text statistics but breaking semantic order; (b) *Plain-Continuation*—rationale fields removed, continuing only on comment text and labels. Table VI summarizes results alongside HARE and Vanilla.

HARE outperforms Shuffled-Rationale by +3.07% F1-Micro (seed 42) and +2.54% (seed 456), providing two-seed evidence that rationale semantic structure contributes beyond extra tokens. Shuffled-Rationale underperforms Vanilla (−0.62%), confirming incoherent rationale adds noise. Plain-Continuation is excluded: the subset lacks Normal examples, causing catastrophic Normal F1 collapse (13.9%).

A. Why Does Boundary Redistribution Occur?

Three non-exclusive mechanisms may explain HARE's disproportionate Offensive gains: **(i) Hate-prototype sharpening**—the Hate-dominant rationale set (60.5%, Table I) may sharpen the Hate prototype, causing marginal examples to be reclassified as Offensive; **(ii) Severity-step reasoning transfer**—the fourth rationale step explicitly trains Hate-vs-Offensive distinction, transferring a finer boundary without inference-time rationale generation; **(iii) Selection-bias artifact**—76.8% rejection removes ambiguous cases, biasing toward high-confidence patterns. The ablation evidence (HARE > Shuffled by +3.07%) supports mechanism (ii). This finding complements English-language observations: Davidson et al. [22] reported ∼40% systematic Offensive→Hate confusion; OLID [23] and HatEval [19] found analogous instability. Our work extends this line of inquiry to Vietnamese, showing the boundary is *malleable via rationale training*—not previously demonstrated.

Table VI
MATCHED-BUDGET ABLATION CONTROLS (SEED 42, %). ‡: NORMAL F1 COLLAPSES TO 13.9% BECAUSE THE 1,221-SAMPLE STAGE 2 TRAINING SUBSET CONTAINS NO NORMAL EXAMPLES (CLASS-DISTRIBUTION CONFOUND).

Model	F1-Micro	F1-Macro
HARE (Stage 2)	**61.45**	**33.71**
Qwen Vanilla (Stage 1)	59.00	33.10
Shuffled-Rationale	58.38	29.52
Plain-Continuation‡	36.35	28.90

Table VII
CONSTRAINED DECODING VS. FREE-TEXT GENERATION (%), MEAN OVER SEEDS {42,123,456}. CONSTRAINED DECODING CATASTROPHICALLY DEGRADES ALL METRICS.

	F1-Micro		F1-Macro	
Model	**Free**	**Constr.**	**Free**	**Constr.**
Vanilla	60.04	21.23	32.75	10.02
HARE	57.08	17.21	35.93	8.04

B. Threats to Validity

Key threats include: differential parse-failure rates distorting aggregate metrics (Sec. VII-C); same-family LLM verification (Gemini used for both generation and filtering); limited-seed ablation (Shuffled confirmed on two seeds; Plain-Continuation on one); and Religion-class data scarcity (<30 test samples). Moreover, the matched-budget ablation cannot fully disentangle three confounded factors: (i) rationale semantic content, (ii) implicit hard-example mining from the 23% selection rate, and (iii) Hate-dominant class rebalancing in the Stage 2 subset (Δ=+26.8pp vs. training distribution). A definitive causal claim would require factorial controls varying each factor independently across ≥5 seeds.

C. Constrained Decoding Experiment

To address parse failures without retraining, we tested constrained decoding via logit masking, restricting the model's vocabulary to valid label tokens at each generation step. Table VII reports multi-seed results.

Constrained decoding collapses F1-Micro from 60%/57% to 21%/17% for Vanilla/HARE respectively. The failure arises

because both models generate labels in a variable, context-dependent order; naïve left-to-right logit masking forces premature commitment to specific label tokens. Retraining with a fixed label ordering is needed to make constrained decoding viable.

VIII. Conclusion

We presented a two-stage QLoRA pipeline adapting HARE's rationale-augmented training to Vietnamese targeted hate speech detection on ViTHSD. Multi-seed evaluation reveals that the aggregate F1-Micro ranking (Vanilla 60.04% vs. HARE 57.08%) is confounded by differential parse-failure rates; a paired bootstrap on parseable outputs reverses it (HARE 46.71% vs. Vanilla 44.58%, p=0.004). The consistent per-label finding is an asymmetric boundary shift: HARE gains +6.84 avg F1 on Offensive labels and +2.77 avg F1 on Hate labels across all seeds, with disproportionate Offensive improvement. Precision–recall decomposition shows these Offensive gains are overwhelmingly recall-driven (+29.0pp Individuals, +31.2pp Groups), suggesting rationale training lowers the Offensive prediction threshold rather than improving discrimination. Matched-budget ablation provides directional evidence for rationale semantic structure contributing beyond extra gradient steps. As a preliminary investigation, we emphasize these findings as directional evidence rather than definitive causal claims; the mechanisms proposed in Sec. VII require further validation with larger seed sweeps and cross-family rationale verification.

For Vietnamese content moderation, HARE's recall advantage (+29pp Individuals#Offensive, +31pp Groups#Offensive) makes it suited for first-pass triage where missing implicit hate carries higher cost than false positives.

Broader implications. The boundary shift phenomenon may generalize to any severity-graded hate speech taxonomy (e.g., OLID [23], HateXplain [12]). Our finding that parse-failure rates can reverse model rankings has methodological relevance for generative classification broadly.

Future work should prioritize: (1) constrained generation via retraining with a canonical label ordering to eliminate parse failures, (2) cross-family rationale verification using GPT-4 or Claude to rule out Gemini-specific label priors, (3) class-balanced Plain-Continuation replication to resolve the seed-dependent Normal collapse, and (4) extension to other Vietnamese hate speech datasets (ViHSD, ViHOS) to test generalizability.

Ethics and Reproducibility Statement

Annotators participated on a voluntary basis with the right to withdraw; exposure to hate content was minimized through work-rotation and peer support. We commit to releasing the 1,744 rationale tuples, QLoRA training code, LoRA adapters, and parseable-only evaluation scripts upon acceptance to support reproducibility.

Acknowledgment

The authors thank Tri Pham Ngoc, Hien Tran Dai, Uyen Huong Tran Dang, and Duong Thuy Nguyen Le (University of Information Technology, VNU-HCM) for insightful discussions on rationale-guided implicit hate speech detection for Vietnamese social media, which inspired the direction of this work.

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